MIL-STD-1209A
30 MARCH 1988
SUPERSEDING
MIL-STD-1209
8 MAY 1967

MILITARY STANDARD

INORGANIC SALTS AND COMPOUNDS, TECHNICAL GRADE

(STRONTIUM CARBONATE THROUGH ZIRCONIUM CARBIDE)

AMSC N/A

FSC 6810

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:42Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
Washington, DC 20301

Inorganic Salts and Compounds, Technical Grade
(Strontium Carbonate through Zirconium Carbide)

MIL-STD-1209A

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, Attn: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

FOREWORD

This standard is approved for use by all Departments and Agencies of the Department of Defense in the selection of items for application. It is intended to prevent the entry of unnecessary items (sizes, types, varieties) into the Department of Defense logistics system. This document is not intended to restrict any service in selecting new items resulting from state-of-the-art changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CONTENTS

Paragraph		Page
1.	SCOPE	1
1.1	Coverage	1
1.2	Application	1
1.3	Classification	1
2.	REFERENCED DOCUMENTS	2
2.1	Government documents	2
2.1.1	Specifications, standards, and handbooks	2
2.1.2	Other Government documents, drawings, and publications	2
2.2	Other publications	3
2.3	Order of precedence	4
3.	DEFINITIONS	5
4.	GENERAL REQUIREMENTS	6
4.1	Packaging data and labeling	6
4.2	Hazardous materials information	6
4.3	Safety	6
4.3.1	Personal protective measures	6
4.3.1.1	Respiratory protection	6
4.3.1.2	Skin protection	6
4.3.1.3	Face and eye protection	6
4.3.1.4	Training	6
4.3.1.5	Exercises	7
4.3.2	Storage conditions	7
4.3.2.1	Flammable, combustible, pyrophoric and ignitable materials	7
4.3.2.2	Water-sensitive fire and explosive hazardous materials	7
4.3.2.3	Incompatible materials	8
4.3.3	Chemical hazardous exposure limits	8
4.3.4	Toxicity	8
4.3.4.1	EPA Toxic (T)	8
4.3.4.2	EPA Acute Hazardous Toxicity (H)	8
4.3.4.3	EPA Extraction Procedure Toxicity (EP Toxicity) (E)	8
4.3.4.4	Hazardous toxic constituents	8
4.4	Pollution and disposal	8
4.4.1	Pollution potential	8
4.4.2	Disposal of excess or unserviceable material	8
4.4.3	Disposal and storage of hazardous wastes	9
4.4.3.1	Cleanup of liquid spills	9
4.4.3.2	Ultimate disposal	10
4.4.4	Disclaimer	10
5.	DETAILED REQUIREMENTS	11
5.1	Strontium Carbonate, Technical	11
5.2	Strontium Chloride, Technical	12
5.3	Strontium Chloride, Hexahydrate, Technical	12
5.4	Strontium Nitrate, Technical	13
5.5	Sulfur Monochloride, Technical	15
5.6	Titanium Tetrachloride, Technical	15
5.7	Titanium Trichloride, Technical	16

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CONTENTS (Continued)

Paragraph		Page
5.8	Titanous Sulfate Solution, 20 Percent, Technical	17
5.9	Zinc Borate, Technical	18
5.10	Zinc Carbonate, Basic, Technical	19
5.11	Zinc Chloride, Technical	20
5.12	Zinc Phosphide, Technical	21
5.13	Zirconium Carbide, Powdered, Technical	22
5.14	Zirconium Carbide, Technical	23
6.	NOTES	25
6.1	Intended use	25
6.2	Subject term (key word) listing	25
6.3	Changes from previous issue	25
6.4	Abbreviations	25

TABLES

I.	Strontium carbonate, technical - typical chemical analysis.	11
II.	Strontium chloride, technical - specification and typical chemical analysis	12
III.	Strontium chloride, hexahydrate, technical - typical chemical analysis.	13
IV.	Strontium nitrate, technical - chemical and physical requirements	14
V.	Strontium nitrate, technical - granulation requirements	14
VI.	Sulfur monochloride, technical - chemical requirements.	15
VII.	Titanium tetrachloride, technical - chemical requirement.	16
VIII.	Titanous sulfate solution, 20 percent - chemical requirement.	17
IX.	Zinc borate, technical - chemical requirements.	18
X.	Zinc borate, technical - granulation requirements.	19
XI.	Zinc carbonate, basic, technical - chemical requirements.	19
XII.	Zinc carbonate, basic, technical - granulation requirements.	19
XIII.	Zinc chloride, technical - chemical requirement	20
XIV.	Zinc phosphide, technical - specification and typical chemical analysis.	21
XV.	Zinc phosphide, technical - granulation requirements.	21
XVI.	Zirconium carbide, powdered, technical - chemical requirements.	22
XVII.	Zirconium carbide, powdered, technical - physical properties.	22
XVIII.	Zirconium carbide, technical - chemical and physical requirements	23

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:42Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Coverage. This standard is a presentation of nomenclature, formulas, physical and chemical properties, specification requirements, military and typical commercial uses, safety information, storage information and disposal information for inorganic salts and compounds, technical grade (strontium carbonate through zirconium carbide). This standard does not include all of the items represented by the title or all those items which are commercially available. It does contain items preferred for use in the selection of inorganic salts and compounds, technical grade (strontium carbonate through zirconium carbide) for application by the Department of Defense.

1.2 Application. Inorganic salts and compounds, technical grade (strontium carbonate through zirconium carbide) are used in pyrotechnics, tracer compositions, fireproofing textiles, smoke mixtures, polymerization and as catalysts.

1.3 Classification. The items in this standard are classified on the basis of chemical composition as inorganic salts and compounds, technical grade.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:42Z
Check the source to verify that this is the current version before use.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-2020 - Chemicals, Liquid, Dry and Paste, Packaging of

MILITARY

MIL-S-12060 - Sulfur Monochloride, Technical (Metric)
MIL-Z-12061 - Zinc Carbonate, Basic, Technical
MIL-Z-12063 - Zinc Borate, Technical (Metric)
MIL-S-20322 - Strontium Nitrate, Anhydrous
MIL-T-50059 - Titanous Sulfate Solution, 20 Percent (Metric)
MIL-Z-82674 - Zirconium Carbide, Powdered, Technical
MIL-Z-85500 - Zirconium Carbide

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data Sheets, Preparation And Submission Of.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this standard to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

Title 29 - Department of Labor, Occupational Safety and Health Administration.
Title 40 - Protection of the Environment; Environmental Protection Agency.
Title 49 - Department Of Transportation; Hazardous Materials Regulations.

DEPARTMENT OF DEFENSE

DOD 4145.19-R-1 - Storage And Materials Handling
DOD 4160.21-M - Defense Utilization And Disposal Manual
DOD 6050.5 - DOD Hazardous Materials Information System, Hazardous Item Listing.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:42Z
Check the source to verify that this is the current version before use.

TB MED 502 (DLAM 1000.2) - Occupational And Environmental Health Respiratory Protection Program.
TB MED 506 - Occupational And Environmental Health Occupational Vision.
TM 38-250 - Packaging, Materials Handling-Preparation Of Hazardous Materials For Military Air Shipment

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

Registry of Toxic Effects of Chemical Substances
Recommendation for Environmental Exposure Limits

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of DODISS specified in the solicitation.

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

TLVs® Threshold Limit Values and Biological Exposure Indices Adopted by American Conference of Governmental Industrial Hygienists (ACGIH).

(Application for copies should be addressed to American Conference of Governmental Industrial Hygienists, 6500 Glenway Avenue, Bldg D-7, Cincinnati, OH 45211-4438.)

ASTM

ASTM E 11 - Wire-Cloth Sieves For Testing Purposes

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

National Fire Codes

(Application for copies should be addressed to National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

MINE SAFETY APPLIANCE COMPANY (MSA)

Mine Safety Appliance Company (MSA) Analysis

(Application for copies should be addressed to Mine Safety Appliance Company, 600 Penn Center Boulevard, Pittsburg, PA 15235)

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:42Z
Check the source to verify that this is the current version before use.

(Nongovernment standards are generally available for reference from libraries. They are also distributed among nongovernment standards bodies and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:42Z
Check the source to verify that this is the current version before use.

3. DEFINITIONS

Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

4. GENERAL REQUIREMENTS

4.1 Packaging data and labeling. All chemicals included in this standard shall be packaged in accordance with Federal Specification PPP-C-2020 and all applicable documents referenced therein. Shipping containers shall be labeled in accordance with current Department of Transportation (DOT) Hazardous Materials Regulations applicable to each chemical. When shipping by military aircraft the requirements of TM 38-250 shall apply. In addition, each item shall be packaged and labeled as specified in the applicable contract or order. All labels shall also comply with Hazard Communication Standard, 29 CFR 1910.1200 (f).

4.2 Hazardous materials information. DOD 6050.5, DOD Hazardous Materials Information System (HMIS) acquires, reviews, stores, and disseminates Material Safety Data Sheet (MSDS) information for all hazardous materials used by DOD. The contractual acquisition of a MSDS is accomplished through use of Federal Acquisition Regulation, paragraph 52.223-3, Hazardous Material Identification and Material Safety Data. The MSDS is prepared in accordance with the instructions in FED-STD-313; and shall comply with requirements of Hazard Communication Standard, 29 CFR 1910.1200 (g).

4.3 Safety.

4.3.1 Personal protective measures. The necessary respiratory, eye and skin protection to be used when handling chemicals shall be prescribed by the responsible installation industrial hygiene, medical and safety authorities.

4.3.1.1 Respiratory protection. Respirators, approved by the National Institute for Occupational Safety and Health (NIOSH) or the Mine Safety and Health Administration (MSHA) or by particular respiratory schedules of the Bureau of Mines (BM) for the compounds being used, may be employed for intermittent, nonroutine exposure (i.e, not exceeding 1 hour/day for 1 day/week), when the installation medical authority determines that there are no feasible engineering or work practice controls, during interim periods when engineering controls are being designed and/or installed, during emergencies, or for supplementing other control measures (refer to TB MED 502 or DLAM 1000.2). Ventilation containment, process controls, or other feasible engineering controls shall be adequate to remove hazardous concentrations. Respiratory protection shall not be used in place of feasible engineering controls.

4.3.1.2 Skin protection. Personnel using these compounds shall be provided with and required to use impervious gloves, sleeves, aprons, and boots whenever indicated. Protective creams and ointments commonly known as "barrier creams" may be of value in certain cases. However, barrier creams shall not be used to replace protective clothing. In case of contact with the skin, wash affected areas thoroughly with water. Eye lavages and emergency showers shall be located where there is a potential for direct contact with harmful chemicals.

4.3.1.3 Face and eye protection. Personnel using these compounds shall be provided with and required to wear chemical splash-proof safety goggles. In addition, face shields shall be provided and worn over the goggles if splashing could occur. In case of contact with the eyes, immediately irrigate with copious amounts of water for at least 20-30 minutes, and obtain medical attention. (Refer to TB MED 506.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

4.3.1.4 Training. Employers shall provide employees with training and information including MSDS on all chemical items in their work area, in accordance with 29 CFR 1910.1200 (h), to ensure that employees know potential hazards of the chemicals with which they come in contact and the symptoms of exposure as well as how these chemicals affect the body and bodily functions. Employees shall be adequately trained to render first aid.

4.3.1.5 Exercises. Participation in exercises shall be stressed to demonstrate skills in the use of personal protective equipment and emergency response equipment.

4.3.2 Storage conditions. DOD 4145.19-R-1 describes general storage practices and requirements for hazardous materials in the DOD supply system. Specific requirements provided in the following paragraphs are supplementary in nature and shall be observed in consonance with the DOD storage regulations.

4.3.2.1 Flammable, combustible, pyrophoric and ignitable materials. A flammable material is generally any solid, liquid, vapor or gas that ignites easily and burns rapidly. Combustible materials are generally those that are difficult to ignite and burn slowly. The DOT, in Part 173, Subpart D, Section 173.115 of 49 CFR, defines a flammable liquid as one having a closed cup flash point below 100°F (37.8°C). A combustible liquid is defined, by DOT in the above reference, as one having a closed cup flash point at or above 100°F and below 200°F (93.3°C). A pyrophoric liquid is defined, by DOT in the above reference, as one that ignites spontaneously in dry or moist air at or below 130°F (54.5°C). Materials with flash points of 200°F or higher are to be considered as burnable. The Environmental Protection Agency (EPA), in Part 261, Subpart C, Section 261.21 of 40 CFR*, designates the criteria for flammable and combustible materials and oxidizers that exhibit the characteristic of ignitability (I). Liquids with closed cup flash points of less than 140°F (60°C) are defined by EPA as ignitable. The autoignition point (temperature) of a substance is generally defined as the minimum temperature required to initiate or cause self-sustained combustion in the absence of a spark or flame. Materials that ignite easily under normal industrial conditions are considered to be dangerous fire hazards. Such materials shall be stored in a manner to prevent ignition and combustion. Easily ignitable substances, such as reducing agents, shall be kept away from strong oxidizing agents. All containers shall be tightly sealed. It is important to provide adequate ventilation in storage areas, and to locate the storage areas of these items away from fire hazards. Ample fire-control equipment shall be easily accessible. Storage buildings, rooms and cabinets shall comply with provisions of the National Fire Codes. The building shall be electrically grounded and signs posted to prevent the lighting of matches or smoking in the area. Flammable storage areas shall be equipped with smoke or fire detection equipment.

4.3.2.2 Water-sensitive fire and explosive hazardous materials. These are materials that react on contact with water or steam to ignite or evolve heat or explosive gases. Such materials exhibit the characteristic of reactivity (R) as designated by the EPA in Section 261.23 of the above reference.* These materials shall be stored in well-ventilated, cool, dry areas. All containers shall be tightly sealed. These materials are a fire hazard in contact with

*Refers only to materials that have become waste materials.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:42Z
Check the source to verify that this is the current version before use.

water or moisture; therefore, it is essential that no sprinkler be used. Otherwise, the building shall conform to that required for storage of flammable materials. The building shall be waterproof, located on high ground, and separated from other storage areas.

4.3.2.3 Incompatible materials. Materials that are chemically incompatible shall be segregated in the storage of both serviceable and unserviceable items. The degree of segregation will depend upon DOD 4145.19-R-1 and local supplementary requirements that insure safe storage conditions. Hazardous storage compatibility codes are provided in the HMIS referred to in 4.2.

4.3.3 Chemical hazardous exposure limits. Chemical hazardous exposure limits for airborne concentrations of substances are obtained from the current TLVs® Threshold Limit Values and Biological Exposure Induces, adopted by the American Conference of Governmental Industrial Hygienists (ACGIH); current Occupational Safety and Health Administration (OSHA) Permissible Exposure Limits (PEL), 29 CFR, Part 1910, Subpart Z; and NIOSH Recommendation for Environmental Exposure Limits. Such information is also shown in MSDSs and the HMIS referred to in 4.2. The identity of sources establishing if a chemical is a carcinogen or potential carcinogen, for hazard communication purposes, is shown in 29 CFR 1910.1200 (d)(4). The more stringent standard shall apply when there is a conflict between standards.

4.3.4 Toxicity. Toxicity information for chemical compounds is available from various publications and from MSDSs, which are collected in DOD 6050.5 Hazardous Materials Information System.

4.3.4.1 EPA Toxic (T). Some chemical compounds have been designated by the EPA as toxic (T) in accordance with the criteria shown in Part 261, Subpart B, Section 261.11(a)(3) of 40 CFR.* Some commercial chemical products are listed as toxic under Subpart D, Section 261.33(f).

4.3.4.2 EPA Acute Hazardous Toxicity (H). Some chemical compounds have been designated by the EPA as acute hazardous (H) in toxicity in accordance with the criteria shown in Subpart B, Section 261.11(a)(2) of 40 CFR.* Some commercial chemical products are listed as acute hazardous in toxicity under Subpart D, Section 261.33(e).

4.3.4.3 EPA Extraction Procedure Toxicity (EP Toxicity) (E). Some chemical contaminants, that are not listed in Subpart D, have been designated by the EPA as exhibiting the characteristic of EP toxicity, and are listed in Section 261.24 Table 1 of 40 CFR.*

4.3.4.4 Hazardous toxic constituents. A list of chemical compounds and substances, shown to have toxic effects on humans or other life forms, is contained in Appendix VIII to 40 CFR Part 261; and the Registry of Toxic Effects of Chemical Substances.

4.4 Pollution and disposal.

4.4.1 Pollution potential. All items described in this standard shall be assumed to have a pollution potential. However, to minimize this potential, the proper use, storage and disposal methods shall be strictly followed.

*Refers only to materials that have become waste materials.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:42Z
Check the source to verify that this is the current version before use.

4.4.2 Disposal of excess or unserviceable material. To minimize disposal problems, it is recommended that no more than a one year's supply of each item listed in this standard be stocked. When stocks have been declared excess or unserviceable, they will be disposed of in accordance with the Defense Utilization and Disposal Manual, DOD 4160.21-M, and applicable DOD Policy Memoranda. Guidance can be obtained from your servicing Defense Reutilization and Marketing Office (DRMO) on procedures required for proper reporting and turn-in.

4.4.3 Disposal and storage of hazardous wastes. Items are classified and managed as hazardous wastes as defined by the Resource Conservation and Recovery Act (RCRA) (Public Law 94-580). Items have been identified as meeting the characteristics (i.e., ignitable, corrosive, reactive or EP toxic) or are listed (i.e., toxic or acute hazardous) according to Identification and Listing of Hazardous Waste, Part 261; 40 CFR; or have been determined to be hazardous wastes by declaration of the Defense Reutilization and Marketing Service (DRMS) in accordance with procedures set forth in DOD 4160.21-M. Disposal of such items shall be managed in accordance with the Installation Environmental Office, the DRMO, or the Safety and Health Office to insure proper reporting of disposal and treatment actions to the US EPA and State; and shall be managed in accordance with Federal, State and local laws. The three main disposal methods are turn-in to the DRMO, on-post disposal by installation personnel, or disposal by commercial contract. Hazardous wastes that cannot be used, or disposed of as stated in 4.4.3.2, shall be stored under environmentally safe conditions until suitable methods of disposal are determined. Short-term storage (less than 90 days) requires proper containment (i.e., packaging and facilities) in accordance with Section 262.34, Part 262 of the above reference. Long-term storage (greater than 90 days) requires permitting by the EPA or by the state under Public Law 94-580 (RCRA), in compliance with the requirements of 40 CFR Parts 264 and 265. Physical custody will be accomplished by the activity with conforming storage or most nearly conforming storage. When physical custody is in question, the Post Commander will make the final decision. In all cases where the wastes are to be collected, stored, transported and disposed of at a state or local permitted disposal facility, the identity and description of the waste shall be maintained and recorded in accordance with Part 262 of the above reference. Transportation of the waste shall be in accordance with Part 263 of the above reference, Standards Applicable to Transporters of Hazardous Waste.

4.4.3.1 Cleanup of liquid spills. To control the migration of spilled or leaking liquids, dike around the item with an inert, dry absorbent (e.g., clay or vermiculite) or follow installations spill plans (Spill Prevention Control and Countermeasure Plan and Installations Spill Contingency Plan). Control entry to the spill site and segregate salvageable materials away from the spill area. Initiate waste cleanup operations immediately in accordance with local procedures. The residue shall be safely handled and transported to an approved or permitted disposal or storage facility. Packaging, labeling, transportation and record-keeping requirements for this waste material are determined by the appropriate Federal and State agencies and local procedures. It is recommended that all activities involving disposal preparation and transportation to commercial facilities be properly coordinated with the appropriate Federal and State agencies responsible for health and environmental aspects of hazardous materials. It is imperative that the proper description of waste

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

accompany the packaged item at all times. Final disposal of the waste item shall be accomplished by reutilization, transfer, donation or sales by DRMS in accordance with DOD 4160.21-M or by ultimate disposal as described in 4.4.3.2. Spill residue, including contaminants, to be turned in to the DRMO shall first be properly identified, containerized, and labeled. For large scale spills that grossly contaminate the environment, the Chemical Transportation Emergency Center (CHEMTREC), can be called for assistance. Applicable procedures of the local spill control plan shall be followed. Necessary respiratory, eye, and skin protection measures are to be used while performing cleanup operations.

4.4.3.2 Ultimate disposal. Ultimate disposal shall be accomplished at a permitted or approved hazardous waste treatment or disposal facility designated by the Installation Environmental Office, DRMO, or Safety and Health Offices.

4.4.4 DISCLAIMER. RECOMMENDED DISPOSAL INSTRUCTIONS ARE FORMULATED FOR USE BY ELEMENTS OF THE DEPARTMENT OF DEFENSE. THE UNITED STATES OF AMERICA IN NO MANNER WHATSOEVER EITHER EXPLICITLY OR IMPLICITLY WARRANTS, STATES, OR INTENDS SAID INSTRUCTION, TO HAVE ANY APPLICATION, USE OR VIABILITY BY OR TO ANY PERSON OR PERSONS CONTRACTING OUTSIDE THE DEPARTMENT OF DEFENSE OR ANY PERSON OR PERSONS CONTRACTING WITH ANY INSTRUMENTALITY OF THE UNITED STATES OF AMERICA AND DISCLAIMS ALL LIABILITY FOR SUCH USE. ANY PERSON USING THESE INSTRUCTIONS WHO IS NOT A MILITARY OR CIVILIAN EMPLOYEE OF THE UNITED STATES OF AMERICA SHOULD SEEK COMPETENT PROFESSIONAL ADVICE TO VERIFY AND ASSUME RESPONSIBILITY FOR THE SUITABILITY OF THESE INSTRUCTIONS TO THEIR PARTICULAR SITUATION REGARDLESS OF SIMILARITY TO A CORRESPONDING DEPARTMENT OF DEFENSE OR OTHER GOVERNMENT SITUATION.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

5. DETAILED REQUIREMENTS

5.1 Name. Strontium Carbonate, Technical $SrCO_3$ FW 147.63
Natural strontionite

5.1.1 Technical description. Strontium carbonate, (pure) exists as colorless, rhombic crystals or white powder, transforms to hexagonal crystals at 926°C. It has refractive indices of 1.516, 1.664 and 1.666 with a density of 3.70 g/cm^3. It loses CO_2 at 1340°C and has a melting point of 1497°C at 69 atmospheres. Its solubility in water is 0.0011 g/100 cm^3 at 18°C and 0.065 g/100 cm^3 at 100°C. It is soluble in solutions of ammonium salts and acids with deco position.

5.1.2 Specification. Manufacturer's requirements (No Government specification).

5.1.2.1 Requirements. Strontium carbonate, technical is available commercially with a minimum assay of 92.0%. The typical chemical analysis is shown in Table I.

TABLE I. Strontium carbonate, technical - typical chemical analysis.

Requirement	% by Wt
Assay (SrO_3)	92.34
Barium carbonate ($BaCO_3$)	1.32
Calcium carbonate ($CaCO_3$)	3.25
Insoluble in HCl	2.16
Aluminum oxide (Al_2O_3)	None
Moisture	0.04

5.1.3 Use. Strontium carbonate, technical is used in pyrotechnics and in the manufacture of iridescent glass.

5.1.4 Safety. Strontium carbonate has a low toxicity and is nonflammable. Overexposure may cause slight irritation of eyes, skin and mucous membranes. Inhalation of dusts or mists shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Strontium carbonate shall be used with adequate ventilation to keep airborne dust at a minimum and below currently adopted maximum exposure level for nuisance particulates.

Strontium carbonate is unstable in the presence of strong acids. Upon decomposition, it emits carbon dioxide.

5.1.5 Storage. Strontium carbonate shall be stored in a cool, dry, well ventilated place in tightly closed containers away from acids and acid fumes.

5.1.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

An EPA Hazardous Waste Classification is not listed in 40 CFR.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

5.2 Name. Strontium Chloride, Technical $SrCl_2$ FW 158.53

5.2.1 Technical description. Strontium chloride, (pure) exists as colorless cubic crystals with a refractive index of 1.650 at 25°C and a density of 3.052 g/cm^3. Its melting point is 875°C and boiling point is 1250°C. Its solubility in water is 53.8 g/100 cm^3 at 20°C and 100.8 g/100 cm^3 at 100°C. It is very slightly soluble in absolute alcohol and acetone and insoluble in ammonia.

5.2.2 Specification. Manufacturer's requirements (No Government specification).

5.2.2.1 Requirements. Strontium chloride, technical is available commercially with a specification and typical chemical analysis as shown in Table II.

TABLE II. Strontium chloride, technical - specification and typical chemical analysis.

Requirement	% by Wt	
	Specification	Typical Analysis
Assay (as $SrCl_2$)	95.0	99.8
Ferric oxide (Fe_2O_3)	0.02 max	0.002
Barium chloride ($BaCl_2$)	0.5 max	0.20
Calcium chloride ($CaCl_2$)	0.4 max	0.20

5.2.3 Use. Strontium chloride, technical is used in pyrotechnics and manufacture of other strontium salts.

5.2.4 Safety. Strontium chloride, has a low toxicity. It is irritating to the eyes, skin and mucous membranes. Inhalation of dusts or mists shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Strontium carbonate shall be used with adequate ventilation.

If heated to decomposition or on contact with acids or acid fumes, strontium chloride can emit toxic chloride fumes.

5.2.5 Storage. Strontium chloride, shall be stored in a cool, dry, well ventilated place in tightly closed containers away from acids and acid fumes.

5.2.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

An EPA Hazardous Waste Classification is not listed in 40 CFR.

5.3 Name. Strontium Chloride, Hexahydrate, Technical
$SrCl_2 \cdot 6H_2O$ FW 266.62

5.3.1 Technical description. Strontium chloride, hexahydrate (pure) exists as colorless, trigonal crystals with refractive indices of 1.536 and 1.487 and a density of 1.93 g/cm^3. It loses four molecules of water at 60°C and six molecules of water at 100°C. Its water solubility is 106.2 g/100 cm^3 at 0°C and 205.8 g/100 cm^3 at 40°C. Its solubility in alcohol is 3.8 g/100 cm^3 at 40°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

5.3.2 Specification. Manufacturer's requirements (No Government specification).

5.3.2.1 Requirements. Strontium chloride, hexahydrate, technical is available commercially with a typical chemical analysis as shown in Table III.

TABLE III. Strontium chloride, hexahydrate, technical - typical chemical analysis.

Requirement	% by Wt
Assay ($SrCl_2$)	95.0
Ferric oxide (Fe_2O_3)	0.02
Barium chloride ($BaCl_2$)	0.5
Calcium chloride ($CaCl_2$)	0.4

5.3.3 Use. Strontium chloride, hexahydrate, technical is used in pyrotechnics, manufacture of other strontium salts, electronic tubes and paint and varnish manufacture.

5.3.4 Safety. Strontium chloride, hexahydrate has a low toxicity. It is irritating to the eyes, skin and mucous membranes. Inhalation of dusts or mists shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Strontium chloride, hexahydrate shall be used with adequate ventilation.

If heated to decomposition or on contact with acids or acid fumes, strontium chloride can emit toxic chloride fumes.

5.3.5 Storage. Strontium chloride, hexahydrate shall be stored in a cool, dry, well ventilated place in tightly closed containers away from acids and acid fumes.

5.3.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

An EPA Hazardous Waste Classification is not listed in 40 CFR.

5.4 Name. Strontium Nitrate, Technical $Sr(NO_3)_2$ FW 211.63

5.4.1 Technical description. Strontium nitrate (pure) exists as colorless cubic crystals with a density of 2.986 g/cm^3. Its melting point is 570°C. Its solubility in water is 70.9 g at 18°C and 100 g/cm^3 at 90°C. Its solubility in absolute alcohol is 0.012 g/100 cm^3. It is very soluble in NH_3 and slightly soluble in acetone.

5.4.2 Specification. Military, MIL-S-20322, Strontium Nitrate, Anhydrous.

This specification covers two grades of strontium nitrate, technical:
Grade A
Grade B

5.4.2.1 Requirements. The military specification chemical, physical and granulation requirements for strontium nitrate, technical are shown in Table IV and Table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

TABLE IV. Strontium nitrate, technical - chemical and physical requirements.

Requirement	% by Wt	
	Grade A	Grade B
Strontium nitrate (including any barium nitrate), min	99.5	99.5
Moisture, max	0.05	0.05
Hygroscopicity, max	0.05	0.05
Acidity (pH)	6.0 - 7.0	6.0 - 7.0
Water - insoluble matter, max	0.02	0.02
Grit, max	0.02	0.02
Chloride (as Cl), max	0.003	0.003
Ammonium compounds (as NH_3), max	0.01	0.01
Sulfur (total), max	0.01	0.01
Barium, max	0.05	1.5
Calcium, max	0.003	0.003
Magnesium, max	0.003	0.003
Sodium, max	0.003	0.003
Iron, max	0.003	0.003
Bismuth, max	0.003	0.003
Lead, max	0.003	0.003
Copper, max	0.003	0.003

TABLE V. Strontium nitrate, technical - granulation requirements.

U S Standard Sieve 1/	% by Wt	
	Grade A	Grade B
Thru 150 micrometer (No. 100), min	95	--
Thru 300 micrometer (No. 50), min		99.9
Retained 106 micrometer (No. 140, min	--	70.0

1/ Standard sieve designation in accordance with ASTM E 11.

5.4.3 Use. Strontium nitrate, technical, grade A is intended for use in pyrotechnic compositions, such as flares. Grade B is intended for military use in tracer compositions.

5.4.4 Safety. Strontium nitrate is moderately toxic and a strong oxidizer. It is irritating to the eyes, skin and mucous membranes. Inhalation of dusts or mists shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Strontium nitrate shall be used with adequate ventilation.

If heated to decomposition it may emit toxic fumes. Strontium nitrate can react with reducing materials.

5.4.5 Storage. Strontium nitrate shall be stored in a cool, dry, well ventilated place in tightly closed containers away from heat, sparks, open flame and reducing materials. Protect containers against shock.

5.4.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

Strontium nitrate has an EPA Hazardous Waste Classification - Ignitable; Waste Number D001.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

5.5 Name. Sulfur Monochloride, Technical S_2Cl_2 FW 135.03
Sulfur chloride
Sulfur subchloride

5.5.1 Technical description. Sulfur monochloride (pure) exists as a yellowish red liquid with a penetrating odor and a refractive index of 1.666 at 14°C. Its density is 1.678 g/cm^3. Its melting point is -80°C and boiling point is 135.6°C. It is soluble in benzene, ether and carbon disulfide.

5.5.2 Specification. Military, MIL-S-12060, Sulfur Monochloride, Technical (Metric).

5.5.2.1 Requirements. The military specification chemical requirements for sulfur monochloride, technical is shown in Table VI.

TABLE VI. Sulfur monochloride, technical - chemical requirements.

Requirement	% by wt
Sulfur content, min	47.0
Chlorine content, min	52.0

5.5.2.1.1 Specific gravity. Sulfur monochloride shall have a specific gravity of 1.680 to 1.690 g/cm^3 (H_2O=1) at 20° _ 1°C.

5.5.3 Use. Sulfur monochloride is intended for use in the manufacture of chemical agent. Commercially it is used in the manufacture of sulfur dyes, insecticides and as a polymerization catalyst for vegetable oils.

5.5.4 Safety. Sulfur monochloride is a corrosive, combustible liquid and very irritating to the eyes, skin and mucous membranes. Inhalation of vapors shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Sulfur monochloride shall be used with adequate ventilation to keep airborne concentrations at or below currently adopted maximum exposure level.

Sulfur monochloride is decomposed by water yielding sulfur, hydrogen sulfide, sulfite and thiosulfate. In acid solution pentathionic and other polythionic acids are formed. If heated to decomposition sulfur monochloride emits toxic fumes of chloride and sulfur oxides.

5.5.5 Storage. Sulfur monochloride shall be stored in a cool, dry, well ventilated place in tightly closed containers.

5.5.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

Sulfur monochloride has an EPA Hazardous Waste Classification - Corrosive; Waste Number D002.

5.6 Name. Titanium Tetrachloride, Technical $TiCl_4$ FW 189.71
Titanic chloride
Titanium (IV) chloride

5.6.1 Technical description. Titanium tetrachloride (pure) exists as a light yellow liquid with a refractive index of 1.61 and a density of 1.726

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

g/cm^3. Its melting point is -25°C and boiling point is 136.4°C. It is soluble in water with hydrolysis, forming hydrochloric acid. It is soluble in alcohol.

5.6.2 Specification. Manufacturer's requirements. (No Government specification).

5.6.2.1 Requirements. Titanium tetrachloride, technical is available commercially with chemical requirements as shown in Table VII.

TABLE VII. Titanium tetrachloride, technical - chemical requirement.

Requirement	% by Wt
Assay ($TiCl_4$), min	95

5.6.2.1.1 Appearance. The material shall be a colorless to reddish-brown liquid, free from sediment.

5.6.2.1.2 Specific gravity. The specific gravity shall not be less than 1.71 or more than 1.74 at 20°/4°C.

5.6.3 Use. Titanium tetrachloride, technical, is intended for use in pyrotechnic compositions. Commercially it is used for the manufacture of iridescent glass and polymerization catalyst.

5.6.4 Safety. Titanium tetrachloride is highly toxic and irritating. Inhalation may cause injury to the upper respiratory tract and acute bronchitis. Skin exposure may cause irritation and burns, and even brief contact with the eyes may cause conjunctivitis and keratitis followed by clouding of cornea. Inhalation of titanium tetrachloride vapors or mists shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Titanium tetrachloride shall be used with adequate ventilation.

Titanium tetrachloride reacts violently with water with the evolution of heat. It fumes strongly when exposed to moist air forming a dense and persistent white cloud.

5.6.5 Storage. Titanium tetrachloride shall be stored in a cool, dry, well ventilated place in tightly closed containers.

5.6.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

Titanium tetrachloride has an EPA Hazardous Waste Classification - Corrosive, Waste Number D002; Reactive, Waste number D003.

5.7 Name. Titanium Trichloride, Technical $TiCl_3$ FW 154.26
Titanous chloride
Titanium (III) chloride

5.7.1 Technical description. Titanium trichloride (pure) exists as dark violet, deliquescent crystals with a density of 2.64 g/cm^3. It decomposes at 440°C. It is soluble in cold and hot water with hydrolysis and the evolution of heat. It is soluble in alcohol and slightly soluble in chloroform. It is insoluble in ether and hydrocarbons.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:42Z
Check the source to verify that this is the current version before use.

5.7.2 Specification. Manufacturer's requirements (No Government specification).

5.7.2.1 Requirements. Titanium trichloride, technical is usually available commercially, as solutions. They are available from one supplier as a 13% solution in approximately 20% hydrochloric acid with a density of 1.199 g/cm^3 at 20°/4°C. It is also available as 20 wt % solution in aqueous hydrochloric acid.

5.7.3 Use. Titanium trichloride, technical is used as a reducing agent and as a cocatalyst in the polymerization of polyolefins. It is also used to remove stains, etc. (stripper) in laundering.

5.7.4 Safety. Titanium trichloride in solution is a strong reducing agent and corrosive. The solid titanium trichloride is flammable. Inhalation of dusts or mists may cause injury to the upper respiratory tract and acute bronchitis. Skin exposure to the solid or solution may cause irritation and burns, and even brief contact with the eyes may cause conjunctivitis followed by clouding of the cornea. Inhalation of titanium trichloride dust or mists shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Titanium trichloride shall be used with adequate ventilation.

Titanium trichloride decomposes in the presence of air and water with the evolution of heat. It is a fire risk in the presence of oxidizing materials.

5.7.5 Storage. Titanium trichloride shall be stored in a cool, dry, well ventilated place in tightly closed containers away from oxidizing materials.

5.7.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

Titanium trichloride has an EPA Hazardous Waste Classification - Corrosive, Waste Number D002.

5.8 Name. Titanous Sulfate Solution, 20 Percent, Technical
Titanium sesquisulfate $Ti_2(SO_4)_3$ FW 383.98

5.8.1 Technical description. Titanous sulfate (pure) exists as a green crystalline powder which is insoluble in cold or hot water, alcohol, ether, or concentrated sulfuric acid. it is soluble in dilute hydrochloric or sulfuric acid.

5.8.2 Specification. Military, MIL-T-50059, Titanous Sulfate Solution, 20 Percent (Metric).

5.8.2.1 Requirements. The military specification chemical requirement for titanous sulfate solution, 20 percent is shown in Table VIII.

TABLE VIII. Titanous sulfate solution, 20 percent - chemical requirement.

Requirement	% by Wt
Assay ($Ti_2(SO_4)_3$, min-max	19 - 22

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

5.8.2.1.1 Stabilizer. Titanous sulfate solution shall contain sufficient stabilizer (citric acid or sodium citrate) to prevent hydrolysis of the titanous sulfate when diluted for use.

5.8.3 Use. Titanous sulfate solution is intended for use as a stripping agent in laundries.

5.8.4 Safety. Titanous sulfate solution is a strong reducing agent and corrosive. It is irritating to the eyes, skin and mucous membranes. Inhalation of vapor or mists shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Titanous sulfate solution shall be used with adequate ventilation.

Titanous sulfate can react with oxidizing materials.

5.8.5 Storage. Titanous sulfate solution shall be stored in a cool, dry, well ventilated place in tightly closed containers away from oxidizing materials.

5.8.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

Titanous sulfate solution has an EPA Hazardous Waste Classification - Corrosive, Waste Number D002.

5.9 Name. Zinc Borate, Technical $3ZnO \cdot 2B_2O_3$ FW 383.35

5.9.1 Technical description. Zinc borate (pure) exists as white triclinic crystals or amorphous powder. The crystals have a density of 4.22 g/cm$_3$ and the powder has a density of 3.64 g/cm^3. Its melting point is 980°C. It is soluble in cold water. The crystalline form is insoluble in hydrochloric acid; the amorphous form is soluble in hydrochloric acid.

5.9.2 Specification. Military, MIL-Z-12063, Zinc Borate, Technical (Metric).

5.9.2.1 Requirements. The military specification chemical requirements for zinc borate, technical is shown in Table IX. The granulation requirements are shown in Table X.

TABLE IX. Zinc borate, technical - chemical requirements.

Requirement	% by Wt	
	Min	Max
Zinc (as ZnO)	---	46.0
Boron (as B_2O_3)	32.0	---
Loss in weight at 110°C	---	0.5
Loss in weight at 600°C	21.0	--
Sulfur (as SO_3)	---	0.5

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:42Z
Check the source to verify that this is the currentversion before use.

TABLE X. Zinc borate, technical - granulation requirements.

U S Standard Sieve 1/	% by Wt	
	Min	Max
Passing 53 micrometer (No. 270)	99.5	--
Passing 45 micrometer (No. 325)	99.0	--

1/ Standard sieve designation in accordance with ASTM E 11.

5.9.3 Use. Zinc borate, technical is intended for use in smoke mixtures. Commercially it is used in fireproofing textiles; fungistat and mildew inhibitor.

5.9.4 Safety. Zinc borate is nonflammable and has a low toxicity. It presents little immediate health hazard. It may cause some irritation to the eyes. Zinc borate shall be used with ordinary precautions and cleanliness. Zinc borate shall be used with adequate ventilation.

If heated to decomposition zinc borate may emit toxic fumes.

5.9.5 Storage. Zinc borate shall be stored in a cool, dry, well ventilated place in tightly closed containers.

5.9.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

An EPA Hazardous Waste Classification is not listed in 40 CFR.

5.10 Name. Zinc Carbonate, Basic, Technical $[ZnCO_3 \cdot xZn(OH)_2]$
Zinc hydroxide carbonate

5.10.1 Technical description. No technical description available due to variability in composition of the ratio of $ZnCO_3$ to $Zn(OH)_2$.

5.10.2 Specification. Military, MIL-Z-12061, Zinc Carbonate, Basic, Technical.

5.10.2.1 Requirements. The military specification chemical requirements for zinc carbonate, basic, technical is shown in Table XI. The granulation requirement is shown in Table XII.

TABLE XI. Zinc carbonate, basic, technical - chemical requirements.

Requirement	% by Wt	
	Min	Max
Assay (as Zn)	54.0	64.0
Loss on ignition	20.0	28.0
Soluble alkali (as Na_2CO_3)	---	0.25
Cadmium (Cd)	---	0.10
Water (uncombined)	---	3.0

TABLE XII. Zinc carbonate, basic, technical - granulation requirements.

U S Standard Sieve 1/	Min
Passing 45 micrometer (No. 325), % by wt	99.0

1/ Standard sieve designation in accordance with ASTM E 11.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:42Z
Check the source to verify that this is the current version before use.

5.10.2.1.1 Appearance. Zinc carbonate, basic, technical shall be free from hard lumps, cakes or agglomerates.

5.10.3 Use. Zinc carbonate, basic, technical is intended for use in the manufacture of smoke mixtures.

5.10.4 Safety. Zinc carbonate, basic has a low toxicity. It presents little immediate health hazard. It may cause some irritation to the upper respiratory system. Zinc carbonate, basic shall be used with ordinary precautions and cleanliness. Zinc carbonate, basic shall be used with adequate ventilation.

If heated to decomposition zinc carbonate, basic may emit irritating fumes.

5.10.5 Storage. Zinc carbonate, basic shall be stored in a cool, dry, well ventilated place in tightly closed containers.

5.10.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

An EPA Hazardous Waste Classification is not listed in 40 CFR.

5.11 Name. Zinc Chloride, Technical ZnCl FW 136.28
Butter of zinc

5.11.1 Technical description. Zinc chloride (pure) exists as white hexagonal, deliquescent crystals with refractive indices of 1.681 and 1.713 and a density of 2.91 g/cm^3 at 25°C. Its melting point is 283°C and boiling point is 732°C. Its solubility in water is 432 g/100 cm^3 at 25°C and 615 g/100 cm^3 at 100°C. It is soluble in alcohol and ether and insoluble in ammonia.

5.11.2 Specification. Manufacturer's requirements (No Government specification).

5.11.2.1 Requirements. Zinc chloride, technical is available commercially with a zinc chloride content as shown in Table XIII.

TABLE XIII. Zinc chloride, technical - chemical requirement.

Requirement	% by wt
Assay ($ZnCl_2$), min	94

5.11.3 Use. Zinc chloride, technical is used as a catalyst, dehydrating and condensing agent in organic synthesis, soldering fluxes; burnishing and polishing compounds for steel and electroplating.

5.11.4 Safety. Zinc chloride is a corrosive material. It is irritating to the eyes, skin and mucous membranes. Inhalation of dusts shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Zinc chloride shall be used with adequate ventilation. Solutions of zinc chloride are also irritating to the eyes and skin. Inhalation of mist is irritating to the respiratory tract. The degree of irritation is dependent upon the concentration of the solution and the length of exposure. Airborne concentrations of zinc chloride fume will be kept at or below the currently adopted maximum exposure level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

If heated to decomposition or on contact with acids or acid fumes, zinc chloride can emit irritating and toxic fumes.

5.11.5 Storage. Zinc chloride shall be stored in a cool, dry, well ventilated place in tightly closed containers away from acids and acid fumes.

5.11.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

Zinc chloride has an EPA Hazardous Waste Classification - Corrosive, Waste Number D002.

5.12 Name. Zinc Phosphide, Technical Zn_3P_2 FW 258.06

5.12.1 Technical description. Zinc phosphide (pure) exists as dark gray, tetragonal crystals with a density of 4.55 g/cm^3 at 18°C. Its melting point is >420°C and boiling point is 1100°C. When strongly heated with the exclusion of air it melts and finally sublimes. It is insoluble in water and alcohol; soluble in HCl and H_2SO_4 with evolution of spontaneously flammable phosphine. When kept dry it is quite stable.

5.12.2 Specification. Manufacturer's requirements (No Government specification).

5.12.2.1 Requirements. Zinc phosphide, technical is commercially available with specification and typical chemical analysis as shown in Table XIV. The granulation requirements are shown in Table XV.

TABLE XIV. Zinc phosphide, technical - specification and typical chemical analysis.

	% by Wt	
Requirement	Specification	Typical Analysis
Zinc phosphide (Zn_3P_2)	94.0 min	95.0
Insoluble in HCl	2.0 max	0.8

TABLE XV. Zinc phosphide, technical - granulation requirements.

	% by Wt	
U S Standard Sieve 1/	Specification	Typical Analysis
Passing 150 micrometer (No. 100)	99.5 min	99.8 min
Passing 45 micrometer (No. 325)	2.0 max	0.8

1/ Standard sieve designation in accordance with ASTM E 11.

5.12.3 Use. Zinc phosphide, technical is used as a source for generation of phosphine gas by reaction with acids. It is also used in the manufacture of bait formulations for rodent control.

5.12.4 Safety. Zinc phosphide is toxic and irritating to the eyes, skin and respiratory system. Contact with the eyes, skin and clothing shall be avoided. Zinc phosphide shall be used with adequate ventilation.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:42Z
Check the source to verify that this is the current version before use.

Zinc phosphide reacts violently with oxidizing agents. It produces toxic and flammable phosphine by reaction with water and acids. When heated, zinc phosphide may emit toxic oxides of phosphorous fumes. Do not re-use container. Completely empty the container and rinse several times with clean water.

5.12.5 Storage. Zinc phosphide shall be stored in a cool, dry, well ventilated place in tightly closed containers away from acids and acid fumes and oxidizing materials.

5.12.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

Zinc phosphide has the following EPA Hazardous Waste Classifications - Zinc phosphide (RT), Acute Hazardous Toxic, Waste Number P122; Zinc phosphide (RT) (when present at concentrations greater than 10%), Waste Number P122. Zinc phosphide (T), Toxic (when present at concentrations of 10% or less), Waste Number U249.

5.13 Name. Zirconium Carbide, Powdered, Technical ZrC FW 103.23

5.13.1 Technical description. Zirconium carbide (pure) exists as gray metallic cubic crystals with a density of 6.73 g/cm^3. Its melting point is 3540°C and boiling point is 5100°C. It is insoluble in water and slightly soluble in concentrated sulfuric acid.

5.13.2 Specification. Military, MIL-Z-82674, Zirconium Carbide, Powdered, Technical.

5.13.2.1 Requirements. The military specification chemical requirements are shown in Table XVI. The physical requirements are shown in Table XVII.

TABLE XVI. Zirconium carbide, powdered, technical - chemical requirements.

Requirement	Min	Max
Zirconium (Zr plus Hf)	87.4	88.5
Hafnium (Hf)	--	2.0
Total carbon (C)	10.8	11.9
Free carbon (C)	--	1.5
Iron (Fe)	--	0.05
Oxygen (O)	--	0.6
Nitrogen (N)	--	0.8

TABLE XVII. Zirconium carbide, powdered, technical - physical properties.

Requirement	Min	Max
Apparent density, g/cm^3	1.46	1.83
Average particle diameter, micrometer	1.5	3.0
Volatiles content, % by wt	---	0.10

5.13.3 Use. Zirconium carbide, powdered, technical is intended for use as a propellant ingredient for the Mark 56 Mod 2 Dual Thrust Rocket Motor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

5.13.4 Safety. Zirconium carbide, powdered, may be irritating to the eyes, skin and mucous membranes. Inhalation of dust or powder shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Zirconium carbide shall be used with adequate ventilation to keep airborn concentrations at or below the currently adopted maximum exposure level for zirconium compounds.

Zirconium carbide, powdered, will react with oxidizers. Zirconium carbide powder or dust will ignite spontaneously.

5.13.5 Storage. Zirconium carbide shall be stored in a cool, dry, well ventilated place in tightly closed containers away from oxidizing materials.

5.13.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

Zirconium carbide powdered has an EPA Hazardous Waste Classification - Ignitable, Waste Number D001; Reactive, Waste Number D003.

5.14 Name. Zirconium Carbide, Technical ZrC FW 103.23

5.14.1 Technical description. Zirconium carbide (pure) exists as gray metallic cubic crystals with a density of 6.73 g/cm^3. Its melting point is 3540°C and boiling point is 5100°C. It is insoluble in water and slightly soluble in concentrated sulfuric acid.

5.14.2 Specification. Military, MIL-Z-85500, Zirconium Carbide.

5.14.2.1 Requirements. The military specification chemical and physical requirements are shown in Table XVIII.

TABLE XVIII. Zirconium carbide, technical - chemical and physical requirements.

Requirement	Min	Max
Weight median diameter, micrometers 1/	6.0	10.0
Fisher average particle diameter, micrometers	2.0	5.0
Zirconium plus hafnium, % by wt	87.9	--
Hafnium (Hf), % by wt	--	2.0
Total carbon, % by wt	11.3	--
Free carbon, % by wt	--	0.5
Iron (Fe), % by wt	--	0.05
Oxygen (O)	--	0.6
Nitrogen (N)	--	0.8

1/ MSA analysis (Mine Safety Appliance Company, Pittsburg, PA).

5.14.2.1.1 Stability. The zirconium carbide (when protected from contamination) shall have a storage life of 18 months from date of delivery to the procuring activity.

5.14.2.1.2 Storage life extension. The storage life of an individual lot may be extended for an additional 18 months provided the zirconium carbide, upon retest, successfully meets the requirement for weight mean diameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

5.14.2.1.3 Workmanship. Workmanship shall be such that the zirconium carbide is uniform in appearance, of consistent high quality and free from visible contamination.

5.14.3 Use. Zirconium carbide, technical is intended for use as an ingredient in solid propellant formulations for rocket motors.

5.14.4 Safety. Zirconium carbide may be irritating to the eyes, skin and mucous membranes. Inhalation of dust or powder shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Zirconium carbide shall be used with adequate ventilation to keep airborn concentrations at or below the currently adopted maximum exposure level for zirconium compounds.

Zirconium carbide will react with oxidizers. Zirconium carbide powder or dust will ignite spontaneously.

5.14.5 Storage. Zirconium carbide shall be stored in a cool, dry, well ventilated place in tightly closed containers away from oxidizing materials.

5.14.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

Zirconium carbide has an EPA Hazardous Waste Classification - Ignitable, Waste Number D001; Reactive, Waste Number D003.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T07:42Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. This standard is intended to cite nomenclature, formulas, physical and chemical properties, specification requirements, military and typical commercial uses, safety information, storage information and disposal information for Inorganic Salts and Compounds, Technical Grade (Strontium Carbonate through Zirconium Carbide) preferred for application by the Department of Defense.

6.2 Subject term (key word) listing.

Exposure limits, hazardous chemicals
Hazardous wastes, disposal and storage of
Information, hazardous chemicals
Safety, hazardous chemicals
Strontium Carbonate, Technical
Strontium Chloride, Hexahydrate, Technical
Strontium Chloride, Technical
Strontium Nitrate, Technical
Sulfur Monochloride, Technical
Titanium Tetrachloride, Technical
Titanium Trichloride, Technical
Titanous Sulfate Solution, 20 Percent, Technical
Zinc Borate, Technical
Zinc Carbonate, Basic, Technical
Zinc Chloride, Technical
Zinc Phosphide, Technical
Zirconium Carbide, Powdered, Technical
Zirconium Carbide, Technical

6.3 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.4 Abbreviations. The use of abbreviations shall be in accordance with MIL-STD-12 where applicable. Metric system abbreviations and symbols shall be in accordance with ASTM E 380.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 Alexandria, VA

POSTAGE WILL BE PAID BY

Commander
U.S. Army Chemical Research, Development
and Engineering Center
ATTN: SMCCR-SPT-S
Aberdeen Proving Ground, MD 21010-5423

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STD-1209A	INORGANIC SALTS AND COMPOUNDS, TECHNICAL GRADE (STRONTIUM CARBONATE THROUGH ZIRCONIUM CARBIDE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:42Z
Check the source to verify that this is the current version before use.